Calorimetry lab

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use an energy balance to analyze mixing and heat transfer.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 2.*
$Q=mc\,\Delta T=10500$ J, assuming the supplied energy heats the sample.

02. Worked example: explain the model
Hot loses 10°C worth, cold gains 10°C worth; equal mc makes energy changes equal/opposite.

03. Read and interpret the model
40°C; equal heat capacities, equal temperature changes in opposite directions.

04. Apply the physics
300 J; with cup gain 50 J, cold sample gains 250 J.

05. Explain your reasoning
Preserve raw measurements; delayed reading allows more exchange with surroundings, biasing the mixing result.

06. Review: connect an earlier idea *Review: Unit 4, Day 15.*
144 J; spring energy scales with compression squared.

07. Exit check
No; equilibrium should lie between initial temperatures under stated assumptions. Check reading or unmodeled energy source.

08. STRETCH: test the model
Above 35°C; weighted mean 40°C.

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01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 2.*
$c=Q/(m\,\Delta T)=1000$ J/(kg·°C). No phase change and no unaccounted loss/input.

02. Worked example: explain the model
Hot loses 10°C worth, cold gains 10°C worth; equal mc makes energy changes equal/opposite.

03. Read and interpret the model
40°C; equal heat capacities, equal temperature changes in opposite directions.

04. Apply the physics
$(0.10\times80+0.30\times20)/0.40=35$°C.

05. Explain your reasoning
Preserve raw measurements; delayed reading allows more exchange with surroundings, biasing the mixing result.

06. Review: connect an earlier idea *Review: Unit 4, Day 15.*
$|x|=\sqrt{(2\,U/k)}=0.06$ m; $|F|=24$ N.

07. Exit check
Water gains 8400 J; metal $c=8400/(0.20\times60)=700$ J/(kg·°C).

08. STRETCH: test the model
Water gains 4200 J; metal cools 75°C; $c=4200/(0.1\times75)=560$ J/(kg·°C).